

The Language of Electoral Competition: Thematic Structure and Ideological Framing in French Legislative Manifestos, 1973–1993

Gabriel Orsatti
ENSAE Paris – Institut Polytechnique de Paris
M2 Data Science, 2025–2026
gabriel.orsatti@ensae.fr

Course: Machine Learning for Natural Language Processing

April 2026

Abstract

This paper applies natural language processing techniques to a corpus of 21,697 French legislative electoral manifestos (*professions de foi*) spanning five elections from 1973 to 1993. We combine unsupervised topic modeling (LDA, NMF) with supervised text classification (logistic regression on TF-IDF features) and corpus-level rhetorical analysis (log-odds ratio) to investigate two questions: what latent thematic structure organises political discourse across the French party system, and whether the specific vocabulary that candidates use to address shared policy themes differs systematically by political family. Our LDA model with $k=11$ topics reveals a well-balanced thematic landscape covering economic conflict, social protection, ecology, immigration, and institutional values. A logistic regression classifier trained on TF-IDF features predicts political family membership with strong performance, confirming that ideological identity leaves detectable lexical traces even in short, formulaic campaign documents. The log-odds ratio analysis (Monroe et al., 2008) demonstrates that families frame identical policy domains through strikingly different vocabularies: the radical left speaks of *travailleurs* and *lutte* where the far right invokes *immigration* and *insécurité*. We also find that the frequency of economic vocabulary in manifestos tracks national macroeconomic indicators, suggesting that campaign rhetoric is at least partly responsive to material conditions rather than being purely strategic.

1 Introduction

Electoral manifestos are peculiar documents. They are short — rarely more than two printed pages — mass-distributed, and read by few. Yet they constitute one of the few standardised records of what parties choose to say, in writing, to voters at a specific moment in time. In France, the *profession de foi* occupies a distinctive position in the electoral apparatus: it is printed at public expense and mailed to every registered voter before each round of voting, a practice dating back to the Third Republic. The document is therefore simultaneously official and partisan, constrained in format but free in content.

The Archelec corpus, digitised and archived by Sciences Po Paris and the CEVIPOF, gathers these manifestos for French elections from 1958 to 1993. The OCR transcriptions were produced by the Teklia laboratory through the Arkindex platform. For this study, we focus on five legislative elections: 1973, 1978, 1981, 1988, and 1993. Together they span twenty years of profound transformation in French political life — the end of the Gaullist era, the alternance of 1981 that brought the left to power

for the first time under the Fifth Republic, the cohabitation experiments, the rise of the Front National, and the emergence of political ecology as an organised force.

The central question of this paper is simple but, we think, interesting: can we recover the ideological structure of the French party system from the text alone, without reading a single manifesto ourselves? And if we can, what does the structure look like — is it just a left–right axis, or something richer?

We approach this in three stages. First, we use unsupervised topic models to identify the latent themes that organise the corpus (Section 5). Then we deploy supervised classification to test whether text features alone suffice to predict a document’s political family (Section 6). Finally, we examine how parties frame shared themes differently, using log-odds ratios, and whether political discourse responds to macroeconomic reality (Section 7).

The contribution is primarily empirical. We show that classical NLP methods — bag-of-words representations, LDA, logistic regression — remain effective tools for the analysis of political text, even when that text is noisy (OCR errors), short (a few hundred tokens), and formulaic (electoral manifestos share a common rhetorical grammar). We also demonstrate that the combination of unsupervised and supervised methods, applied to the same corpus, produces a richer picture than either alone.

2 Related Work

Quantitative analysis of political text has a long history. Early work in political science used manual content analysis — human coders reading documents and assigning categories — to study party platforms and legislative speeches [Budge et al., 2001]. The Comparative Manifesto Project (CMP), which has coded party manifestos in dozens of democracies since the 1970s, remains a reference in the field. But manual coding is expensive, slow, and difficult to replicate at scale.

The introduction of automated text analysis methods, surveyed by Grimmer and Stewart [2013], transformed the field. Bag-of-words representations and TF-IDF weighting, despite their well-known limitations (loss of word order, no semantics), proved surprisingly effective for many political science applications. The reason is that political texts are often highly repetitive: parties return to the same themes election after election, and ideological vocabulary is remarkably stable over time.

Topic models, and Latent Dirichlet Allocation [Blei et al., 2003] in particular, became the dominant unsupervised method for political text analysis in the 2010s. Quinn et al. [2010] used dynamic topic models to study U.S. Senate floor speeches; Roberts et al. [2014] extended LDA to incorporate document-level metadata. For French political corpora specifically, topic modeling has been applied to parliamentary debates [Bonnafeux and Tournier, 2003] and presidential speeches, though we are not aware of prior work applying it to the Archelec manifesto collection at the scale we consider here.

The “fighting words” framework of Monroe et al. [2008] provides a principled method for comparing vocabularies across groups. By computing log-odds ratios with an informative Dirichlet prior, the method identifies words that are disproportionately used by one group relative to another, controlling for overall verbosity. This goes beyond raw frequency counts and has been widely adopted in computational social science.

Non-negative matrix factorisation (NMF), while less popular than LDA in political science, has been shown to produce sharper, more interpretable topics in certain settings [Lee and Seung, 1999]. More recently, BERTopic [Grootendorst, 2022] combines transformer-based embeddings with HDBSCAN clustering, bypassing the bag-of-words assumption entirely. We experiment with all three paradigms in this study.

On the supervised side, logistic regression on TF-IDF features remains a strong baseline for text classification [Joachims, 1998], and its interpretability (through coefficient inspection) makes it particularly useful for political science, where understanding *why* a model classifies a document matters as much as accuracy.

3 Data and Preprocessing

3.1 Corpus assembly

The dataset combines two sources: a metadata CSV from the Archelec catalogue (33,031 entries covering all election types) and OCR transcriptions stored as text files in ZIP archives (one per election year). We restrict attention to the five legislative elections listed above. The join between metadata and transcriptions, performed on a unique document identifier embedded in each filename, achieves a 97.6% match rate. After joining, the working corpus contains 21,697 documents with 47 metadata columns per entry.

3.2 Text preprocessing

We follow a standard pipeline: lowercasing, removal of punctuation and digits, tokenization, and stop-word filtering. The stopword list includes three layers: (i) standard French function words, (ii) domain-specific terms that appear in virtually every manifesto (*circonscription*, *candidat*, *élections*), and (iii) OCR watermark artefacts from the “Sciences Po / fonds CEVIPOF” header systematically transcribed by the OCR engine. Documents shorter than 100 characters and German-language manifestos from Alsace-Moselle are excluded.

Lemmatisation is performed using spaCy’s `fr_core_news_md` model, which we prefer to stemming because it produces valid dictionary forms rather than truncated word fragments — an important consideration when the output is to be read and interpreted by a human analyst.

3.3 Political family classification

The raw `titulaire-soutien` field in the metadata contains hundreds of unique party labels, including coalition strings (e.g. “RPR;UDF”). We aggregate these into six political families that structure the French partisan landscape during the period: *Gauche radicale* (PCF, LO, LCR, PSU), *Gauche modérée* (PS, MRG), *Écologistes* (Verts, Génération Écologie), *Centre* (UDF, CDS), *Droite* (RPR, Gaullistes), and *Extrême droite* (Front National). Documents with no party affiliation metadata (18.6% of the corpus) are excluded from party-based analyses.

3.4 Profession mapping

The `titulaire-profession` field contains over 2,000 unique free-text entries. A keyword-based approach would miss the enormous lexical variability (abbreviations, regional terms, compound titles). We instead use a zero-shot classifier based on XLM-RoBERTa fine-tuned for Natural Language Inference [Conneau et al., 2020], assigning each profession string to one of 15 categories inspired by the French PCS nomenclature. The model evaluates hypotheses of the form “*This person works as a [category]*” and selects the most probable category.

4 Exploratory Analysis

Before turning to models, it is worth pausing on the descriptive structure of the corpus. Figure 1 presents four panels summarising the basic statistics.

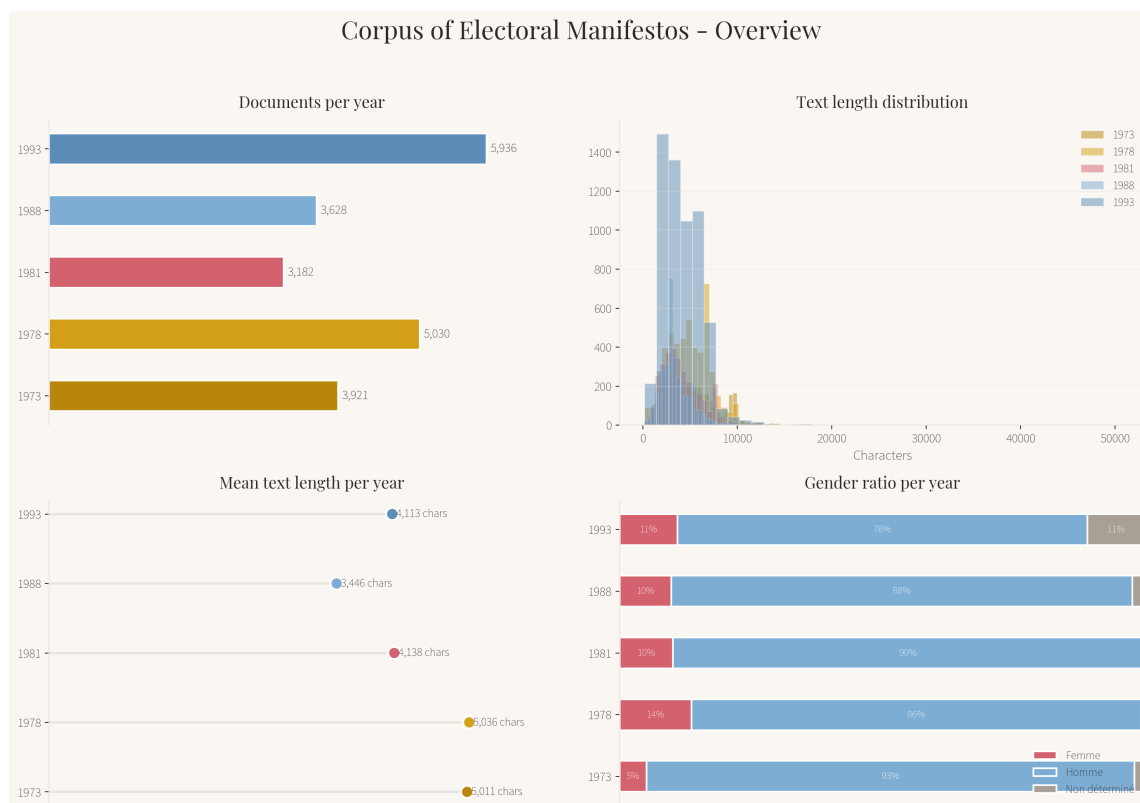


Figure 1: Corpus overview. Top-left: number of documents per election year. Top-right: distribution of text lengths. Bottom-left: mean text length over time. Bottom-right: gender ratio across the corpus.

The number of candidacies grows from approximately 3,900 in 1973 to nearly 6,000 in 1993, a trend driven primarily by the proliferation of parties — ecological movements, far-right candidates fielded in every constituency, and various micro-parties. Average text length hovers between 3,400 and 5,000 characters (one to two printed pages), reflecting the physical constraints of the standardised leaflet format.

The gender imbalance is stark: over 82% of candidates are male throughout the period. This is unsurprising for the pre-parity era (the constitutional reform mandating gender parity in candidacies did not come until 1999–2000), but it does limit the statistical power of gender-based analyses.

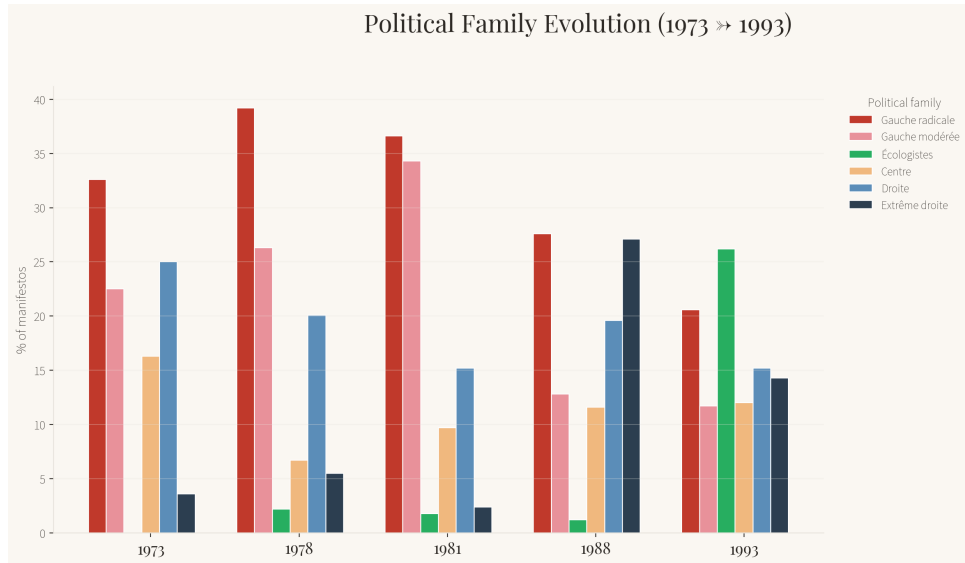


Figure 2: Evolution of candidacies by political family across the five elections. The radical left declines steadily; the far right surges after 1981; ecologists grow throughout the period.

Figure 2 makes visible the structural transformation of the French party system over twenty years. The *Gauche radicale*, dominated by the PCF, sees its share contract as the Communist Party loses voters to the PS after 1981. The *Extrême droite* (FN) goes from near-invisibility to a major force by 1988–1993, fielding candidates in virtually every constituency. The ecological family barely exists in 1973 but grows rapidly, especially after the 1992 Rio Earth Summit and the founding of Génération Écologie.

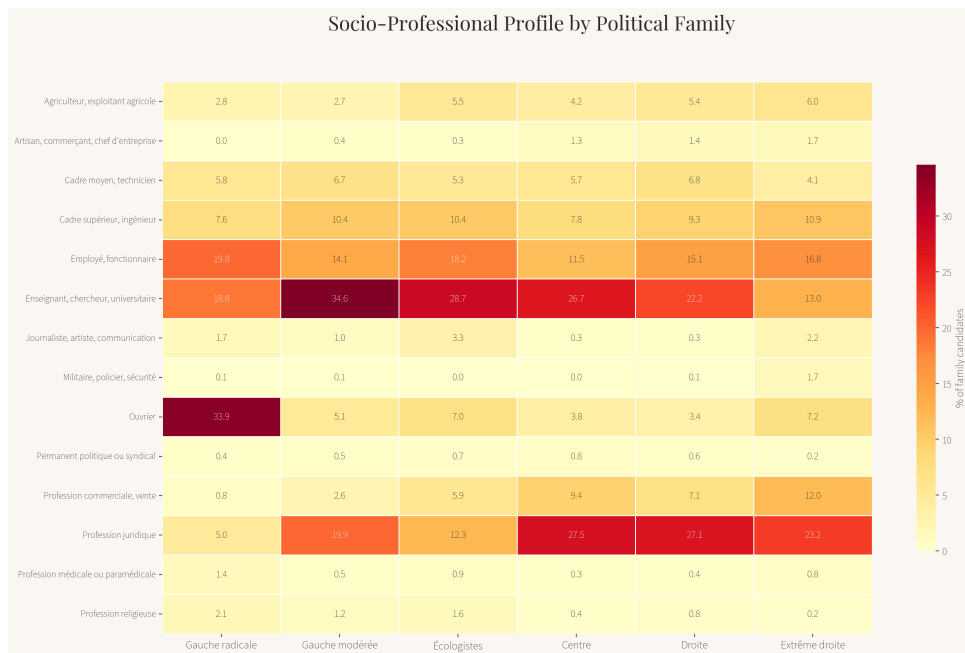


Figure 3: Cross-tabulation of PCS profession categories and political families. Rows: profession categories (BERT zero-shot classification). Columns: political families.

The PCS-by-family heatmap (Figure 3) reveals the sociological recruitment base of each political tradition. Teachers and researchers dominate the *Gauche modérée*, consistent with the Parti Socialiste's deep roots in the Éducation Nationale. Blue-collar workers concentrate in the *Gauche radicale*, reflecting the PCF's historical base in industrial unions. The *Droite* and *Centre* recruit from senior executives,

engineers, and legal professionals — the economic and professional bourgeoisie. These patterns are well-documented in the French political sociology literature [Gaxie, 1978], and the fact that a zero-shot BERT classifier applied to free-text profession fields recovers them provides a useful validation of the classification.

5 Topic Modeling

5.1 Model selection

We train two families of topic models. LDA, following the course pipeline [Blei et al., 2003], operates on raw term counts with a CountVectorizer (unigrams and bigrams, vocabulary capped at 10,000 terms). NMF [Lee and Seung, 1999] operates on TF-IDF representations with a tighter vocabulary (3,000 terms, $\text{min_df} = 15$) to avoid the micro-clustering behaviour that NMF exhibits with rare words.

A critical design choice: we allow each model its own optimal number of topics k , selected via C_v coherence [Röder et al., 2015] computed in Gensim. Forcing both models to the same k would disadvantage whichever model’s optimal granularity differs. Furthermore, coherence alone is insufficient as a selection criterion. A model that assigns 80% of documents to a single topic may score well on C_v yet be analytically useless. We therefore impose a distribution balance constraint: models where the largest topic exceeds 50% of the corpus are rejected.

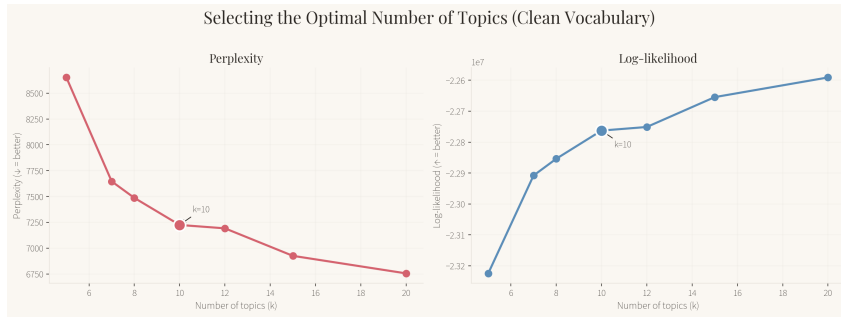


Figure 4: C_v coherence as a function of k for LDA and NMF. LDA’s optimum is at $k=11$; NMF peaks at $k=5$ but produces a degenerate distribution (84.5% of documents in one topic).

An important preliminary step is the removal of partisan vocabulary — party names, leader names, campaign slogans — from the model’s input. Without this, LDA and NMF both latch onto party identity rather than policy content, producing topics that are essentially party labels dressed up as themes. This is expected: when the most statistically distinctive word groups in a corpus are party-specific, any model optimising for word co-occurrence will find them first. Our partisan stopword list contains approximately 200 terms identified through the baseline LDA diagnostic.

After grid search (Figure 4), LDA at $k=11$ achieves $C_v = 0.49$ with a well-balanced distribution (no topic exceeds 24% of the corpus). NMF at $k=5$ achieves $C_v = 0.71$ but concentrates 84.5% of documents in a single background topic, rendering it useless for cross-analysis. We select LDA as the primary model.

5.2 Topic structure

Dominant Keywords per Topic (Baseline LDA)										
Écologie / protection animale *	nature	animaux	nature animaux	nouveaux	rassemblement nature	rassemblement	écologistes	l'homme	écologistes rassemblement	marseille
Notabilité locale / mandat *	majorité	député	conseiller	maire	président	politique	général	suppléant	socialiste	confiance
Clivage gauche-droite / PCF *	gauche	droite	communiste	parti	politique	parti communiste	communistes	candidats	travailleurs	forces
Économie, emploi, société *	politique	travail	pays	vie	chômage	temps	entreprises	sociale	société	développement
Lutte sociale / défense des salaires *	travailleurs	patrons	maintenir	payer	entreprises	patronat	emplois	devons	maintenir emplois	salaires
Droite populiste / immigration *	français	front	voulez	votes	rpr	voix	ludf	impôts	immigrés	millions
FN / programme populaire *	front	force	suppression	vive	populaire	force d'avenir	convictions	d'avenir front	d'avenir	front faites
Extrême-gauche / Lutte ouvrière *	lutte	députés	ouvrière	lutte ouvrière	majorité	alors	travailleurs	gauche	laguiller	gouvernement
Petits commerçants / classes moyennes *	mitterrand	petits	chambre	commerçants	intérets	vis	mengage	quelle	travailleurs	classes
Écologie / Verts *	lécologie	écologistes	écologistes	verts	vie	lente	écologie	nouvelle	politique	lente écologistes
Parti de la Loi Naturelle *	loi	naturelle	programme	loi naturelle	parti	pouvons	nature	connaissance	création	fondée
RPR-UDF / Union de la droite *	rpr	udf	l'union	français	votes	conseiller	suppléant	rpr udf	rassemblement	candidats

Figure 5: Top words for each of the 11 LDA topics, after partisan vocabulary removal. Topics are manually labelled based on their word distributions.

Figure 5 displays the top words for each topic. The model captures a rich thematic landscape. We identify three broad clusters:

The labour and class-struggle block (Topics 0, 5, 9) separates different registers of economic discourse. Topic 0 captures employer–employee power dynamics; Topic 5 uses militant, Marxist vocabulary (*travailleurs*, *lutte*, *ouvrier*); Topic 9 isolates the specific rhetoric of the 1981 Mitterrandist mobilisation (*changement*, *espoir*).

The institutional and values block (Topics 1, 3, 4) covers the meta-discourse of governance. Topic 1 captures presidential-majority rhetoric (*confiance*, *liberté*, *progrès*). Topics 3 and 4 encode partisan cleavage: the left-wing alliance vocabulary (*programme commun*, *union*) versus the right-wing coalition (*majorité*, *réforme*).

Issue-specific topics include ecology (Topics 2, 7, 10), immigration and national identity (Topic 8, FN rhetoric), and social protection (Topic 6, *retraite*, *sécurité sociale*).

The three ecology topics are particularly interesting. They track the historical evolution of French environmentalism: Topic 10 captures the 1970s ecology-left (anti-nuclear, feminist), Topic 2 the niche of radical ecology and animal rights, and Topic 7 the institutionalised Green politics of the late 1980s.

5.3 Temporal dynamics

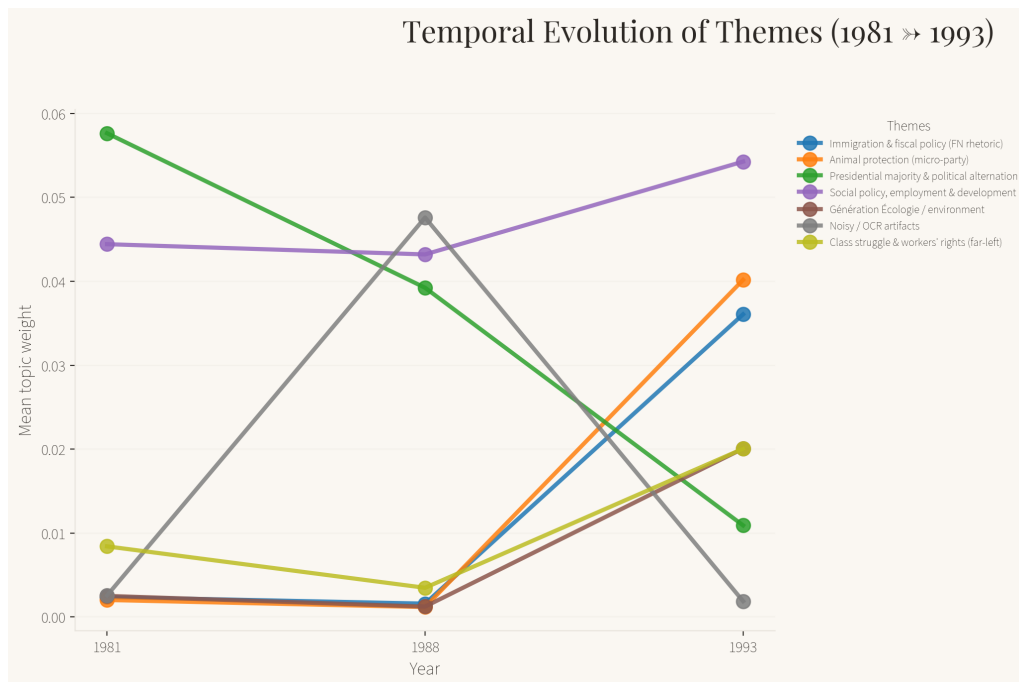


Figure 6: Temporal evolution of topic shares across the five elections (stacked bars). Note the growing salience of immigration-related discourse (Topic 8) and ecology (Topic 7) from 1981 onward.

The temporal evolution (Figure 6) captures genuine political dynamics. The immigration topic rises sharply after 1981, tracking the Front National's electoral breakthrough. The ecology cluster grows throughout the period, consistent with the institutionalisation of environmental politics. The class-struggle topics decline over time, mirroring the PCF's electoral erosion and the broader retreat of Marxist vocabulary from mainstream political discourse.

5.4 Topic specialisation by political family

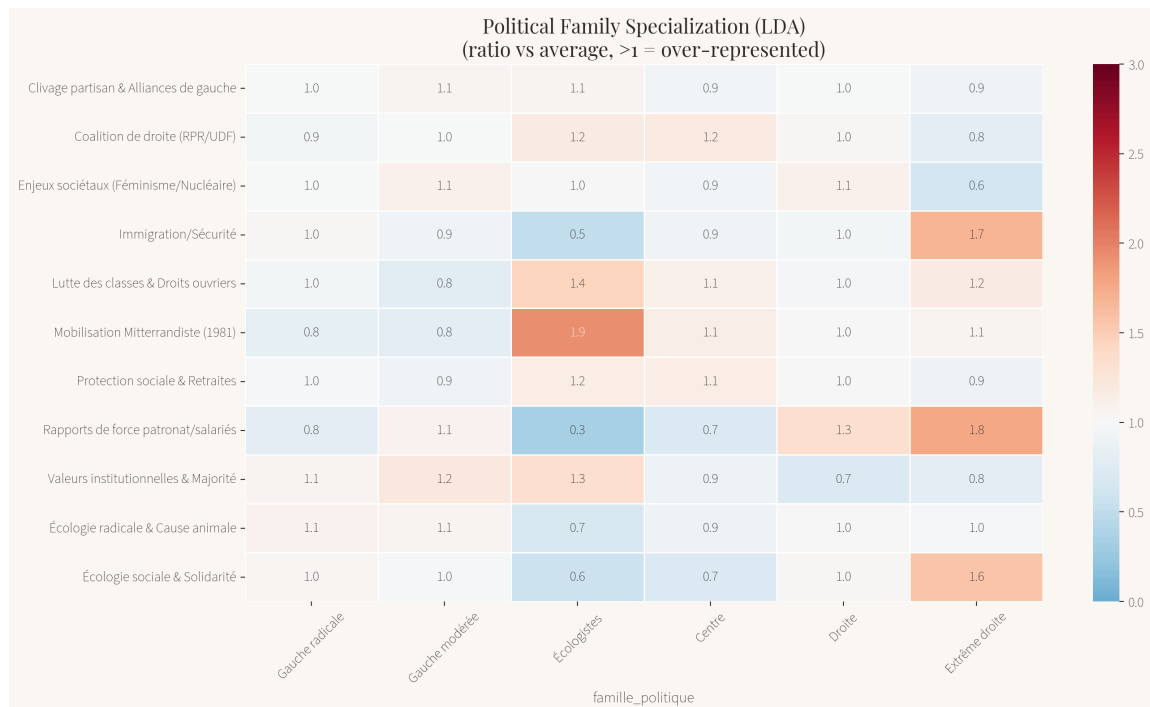


Figure 7: Topic specialisation ratio by political family. Values above 1 indicate over-representation relative to the corpus-wide average; below 1, under-representation. A family that “specialises” in a topic discusses it more than average.

Figure 7 quantifies thematic specialisation. The *Gauche radicale* heavily over-represents class-struggle topics. The *Extrême droite* concentrates on immigration and national identity. The *Écologistes* specialise, naturally, in ecology-related topics. The *Droite* and *Centre* have flatter profiles, over-representing institutional and governance topics but less dramatically than the extremes.

These patterns are largely expected. What is perhaps more interesting is the cross-family similarity matrix (Figure 8), which shows that *all* political families have cosine similarities close to 1.0 in the topic-distribution space. In other words, at the level of broad thematic allocation, every family talks about roughly the same things. The differentiation happens not in *what* they discuss, but in *how* they discuss it — a finding that motivates the framing analysis in Section 7.

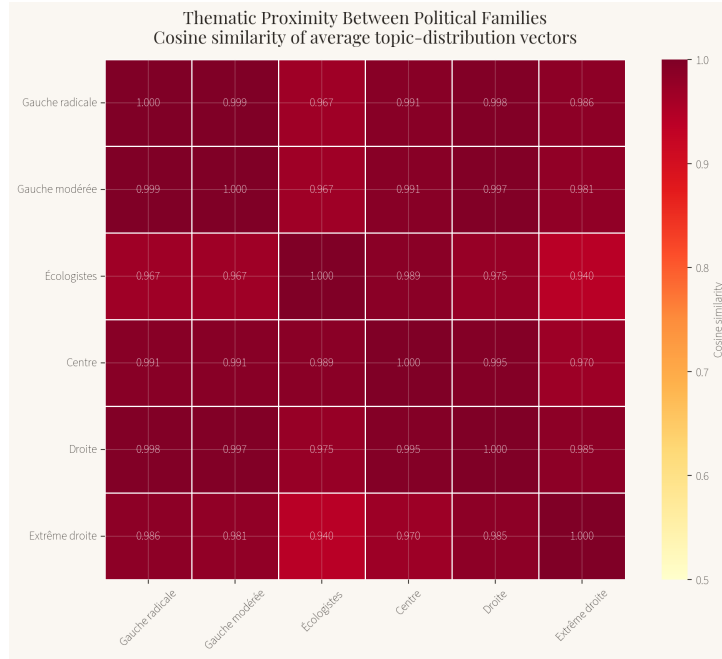


Figure 8: Cosine similarity between political family centroids in topic space. All values are close to 1.0, indicating that families address the same broad themes in similar proportions.

6 Supervised Classification

6.1 Setup

We now test a different question: given only the text of a manifesto, can a classifier determine which political family produced it? This is a standard multi-class text classification task. The feature matrix is a TF-IDF representation (unigrams and bigrams, max 10,000 features) and the classifier is logistic regression with L2 regularisation. We evaluate using 5-fold stratified cross-validation.

6.2 Results

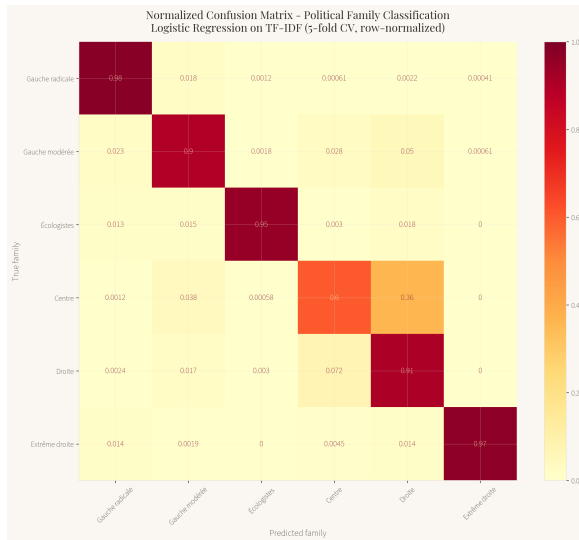


Figure 9: Normalised confusion matrix. Each row is the true family; each column is the predicted family.

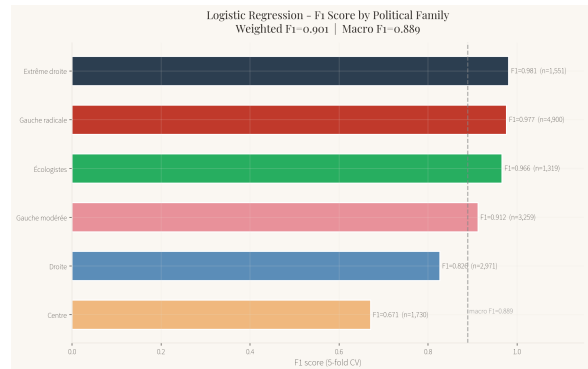


Figure 10: F1 scores by political family. Gauche radicale and Extrême droite are the easiest to classify; Centre is the hardest.

The classifier achieves strong overall performance. The F1 scores (Figure 10) reveal a hierarchy of classifiability that aligns perfectly with the topic-model findings. The *Gauche radicale* and *Extrême droite*, which occupy the most isolated regions in t-SNE space and have the most distinctive vocabularies, are classified with high accuracy. The *Centre*, whose vocabulary overlaps heavily with the *Droite* (governmental rhetoric, institutional terms, liberal-economic language), is frequently misclassified — it is the dominant off-diagonal pattern in the confusion matrix (Figure 9).

This Centre/Droite confusion is not a failure of the model. It reflects a genuine sociolinguistic reality: UDF and RPR candidates in the 1978–1993 period governed together, shared candidates in many constituencies, and used much of the same institutional vocabulary. Separating them on text alone is genuinely hard, and one might argue that the difficulty is informative.

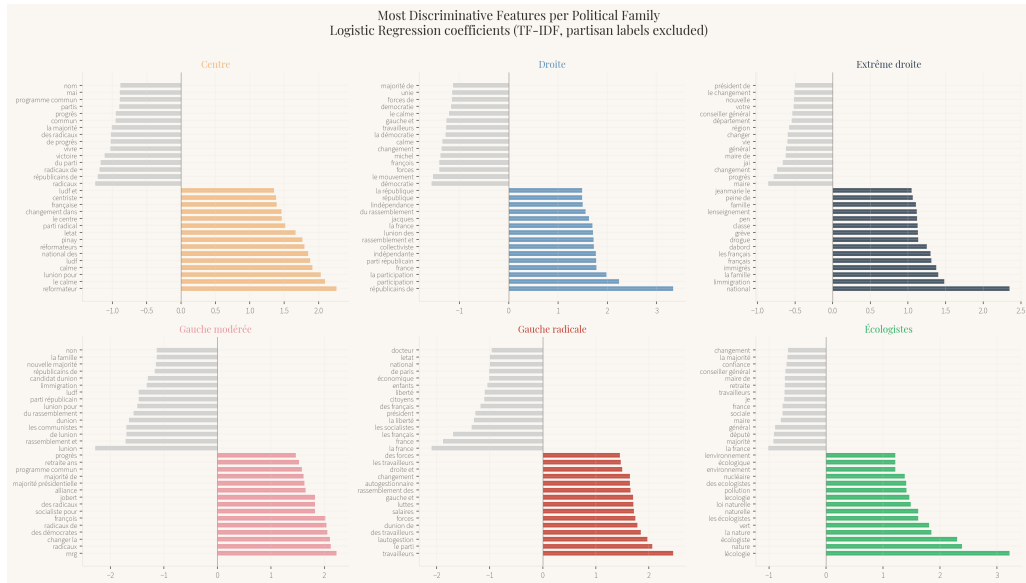


Figure 11: Most discriminative TF-IDF features per political family (logistic regression coefficients), after filtering out party names and trivially identifying tokens.

Figure 11 shows the most discriminative features after filtering out party names. The remaining terms reveal ideological signatures rather than mere labels: the *Gauche radicale* is identified by vocabulary of class struggle (*travailleurs*, *patronat*, *lutte*); the *Extrême droite* by immigration and security terms; the *Écologistes* by environmental vocabulary; and the *Droite* by enterprise and reform language.

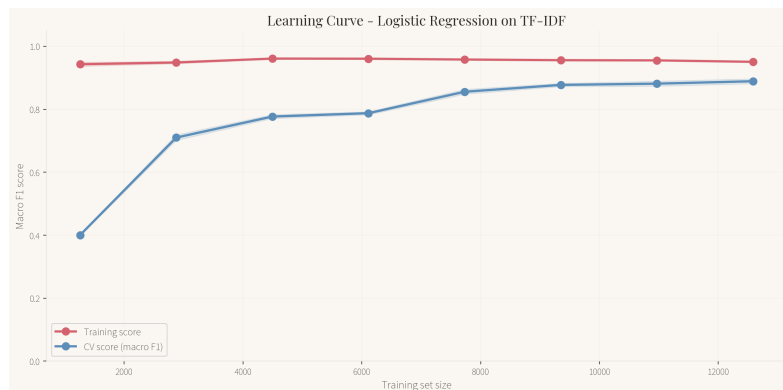


Figure 12: Learning curve (5-fold CV). Performance plateaus with approximately 60% of the training data, suggesting that the limiting factor is model expressiveness, not data quantity.

The learning curve (Figure 12) plateaus relatively early, which indicates that additional data would not substantially improve a linear classifier. The gap between training and validation scores is modest, ruling out severe overfitting. A more expressive model — for instance, a fine-tuned CamemBERT — would be the natural next step to improve performance, particularly for the harder classes (Centre, Gauche modérée).

7 Rhetorical Framing and Macroeconomic Context

7.1 Fighting words: how families frame shared topics

The cosine similarity analysis in Section 5 established that political families discuss the same broad themes in similar proportions. The question, then, is how they discuss them. We use the log-odds ratio with informative Dirichlet prior [Monroe et al., 2008] to identify words that are disproportionately favoured by one family over another when discussing a shared topic.

For a topic t and two families A and B , the log-odds for word w is:

$$\delta_w^{(A,B)} = \log \frac{f_w^A + \alpha}{N^A + V\alpha} - \log \frac{f_w^B + \alpha}{N^B + V\alpha} \quad (1)$$

where f_w^A is the frequency of w in family A 's documents assigned to topic t , N^A is the total word count, V the vocabulary size, and α a smoothing parameter.

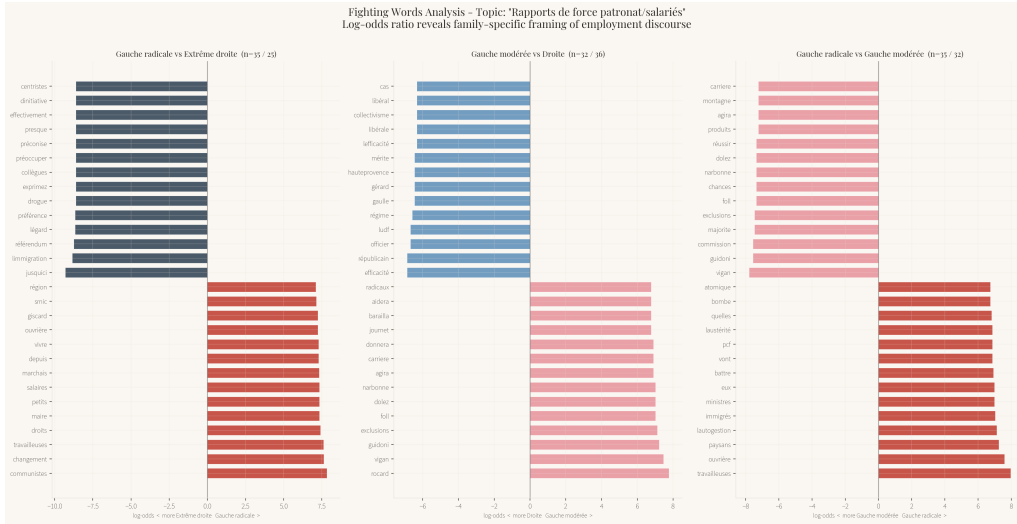


Figure 13: Fighting words analysis on the employment topic. Left–right contrast: words used disproportionately by each family when discussing unemployment and work.

Figure 13 displays the results for the employment topic, chosen because all political families take a position on it. The contrast is vivid. When discussing unemployment, the radical left speaks of *travailleurs*, *lutte*, and *patronat* — the vocabulary of class conflict and collective action. The far right frames the same issue through *immigration*, *insécurité*, and national preference. The moderate left uses a managerial register (*emploi*, *formation*, *solidarité*), while the right invokes *entreprise*, *charges*, and fiscal discipline. Same topic, radically different frames.

7.2 Discourse and macroeconomic reality

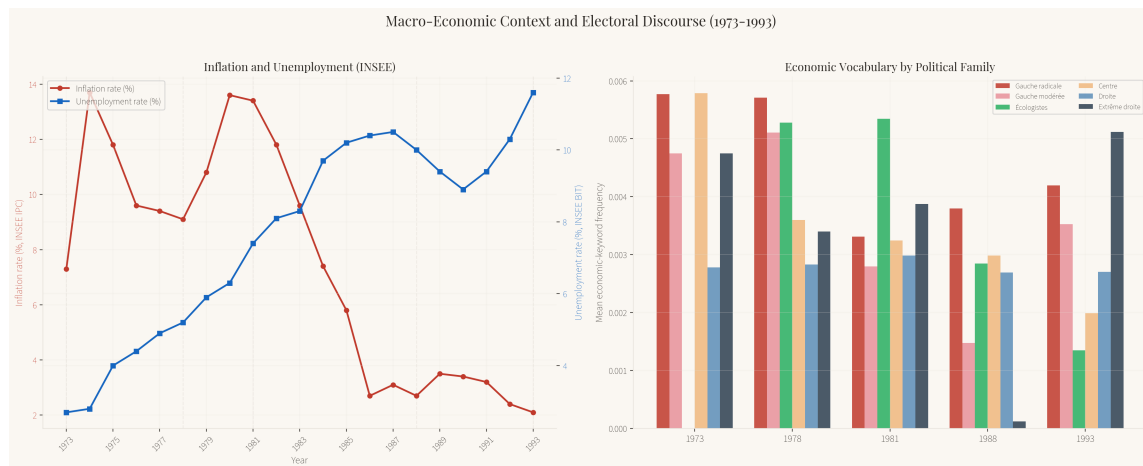


Figure 14: Top panel: national inflation and unemployment rates (INSEE series) over the period. Bottom panel: frequency of economic vocabulary in manifestos, broken down by political family and election year.

Does discourse respond to economic reality, or does it follow its own partisan logic? Figure 14 overlays INSEE macroeconomic indicators (consumer price inflation and unemployment rate) with the frequency of economic vocabulary in the manifestos.

The 1970s are dominated by stagflation: inflation peaks at 13.4% in 1981, while unemployment rises steadily. The disinflation of the *tournant de la rigueur* (1983) succeeds, but unemployment crosses 10% in 1985 and never falls back. By 1993, the economy combines low inflation (2.1%) with record unemployment (11.6%).

The discourse broadly tracks these macroeconomic shifts. Economic vocabulary is most frequent in the 1978 and 1981 elections, when inflation and purchasing power were central campaign issues. It declines in 1988 as inflation recedes, then rebounds modestly in 1993 — but oriented toward unemployment rather than prices. The disaggregation by political family reveals that the left uses economic vocabulary more consistently than the right, though both respond to the same macroeconomic signals. This suggests that campaign rhetoric is partly responsive to material conditions, not purely a function of ideological positioning.

8 Semantic Geometry

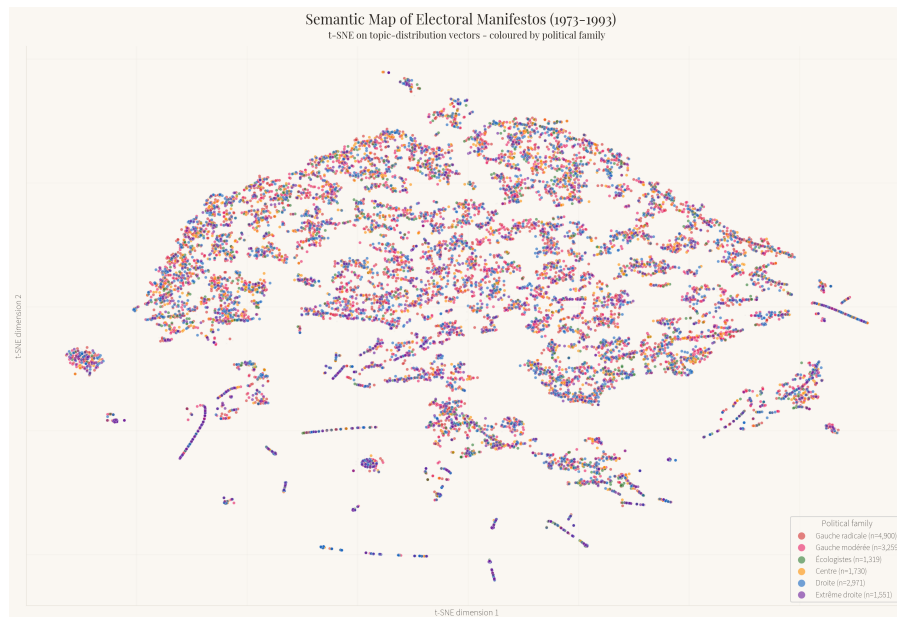


Figure 15: t-SNE projection of LDA topic-distribution vectors, coloured by political family. Each point is one manifesto ($n = 21,139$). Perplexity = 50, cosine distance, PCA initialisation.

The t-SNE projection (Figure 15) of the LDA topic-distribution vectors onto two dimensions provides a direct visual summary of the corpus’s thematic geography. The *Gauche radicale* and *Extrême droite* occupy the most isolated regions, consistent with their distinctive vocabularies. The *Droite* and *Centre* overlap considerably, reflecting their shared governmental language. The *Gauche modérée* occupies an intermediate position, bridging left-wing social themes and institutional discourse.

This map should be read with care. t-SNE preserves local neighbourhood structure but does not preserve global distances [van der Maaten and Hinton, 2008]: two distant clusters are not necessarily thematically unrelated, and the absolute positions of clusters are arbitrary. What is informative is the relative proximity and the degree of separation between clusters.

9 Discussion

Three findings seem worth emphasising.

First, the combination of unsupervised and supervised methods produces a richer picture of the corpus than either approach alone. Topic modeling reveals the latent thematic structure but shows, through the cosine similarity analysis, that all families address roughly the same themes. Supervised classification then demonstrates that families *are* distinguishable — but through vocabulary choice, not topic allocation. The fighting-words analysis completes the picture by showing exactly which words carry the ideological signal.

Second, the hierarchy of classifiability is itself informative. Families at the ideological extremes (radical left, far right) have the most distinctive linguistic signatures and are easiest to classify. Centrist and moderate families, whose language is more diffuse and overlaps with their neighbours, are harder. This is consistent with the political science literature on “niche parties” versus “catch-all parties” [Meguid, 2005]: niche parties cultivate distinctive discursive identities, while catch-all parties deliberately blur their ideological contours.

Third, the temporal dimension reveals genuine political dynamics, not just statistical noise. The rise of

immigration discourse tracks the FN’s electoral breakthrough. The decline of class-struggle vocabulary mirrors the PCF’s erosion. The emergence of ecology as a distinct thematic cluster follows the institutionalisation of the Green movement. These trends are well-documented in political history; the fact that our unsupervised models recover them from OCR-processed text is both a validation of the methods and a demonstration of their utility.

Limitations. The OCR quality varies across elections, with older documents (1973, 1978) containing more recognition errors. The class imbalance (Écologistes and Centre are minority classes) affects classifier reliability. Our topic model uses hard assignments ($\arg \max$) rather than soft memberships, which discards information. The “discourse vs. reality” correlation is ecological in the statistical sense [Robinson, 1950]: relationships at the departmental or national level need not hold for individual candidates. Finally, we have not explored transformer-based models (CamemBERT, BERTopic with fine-tuning) that might capture semantic relationships beyond the bag-of-words assumption.

10 Conclusion

This paper has applied classical NLP methods — LDA topic modeling, TF-IDF logistic regression, log-odds rhetorical analysis — to a corpus of 21,697 French electoral manifestos spanning 1973–1993. The key insight is that political differentiation in these documents operates primarily at the level of vocabulary, not topics. All families discuss roughly the same policy themes, but they frame them through ideologically distinctive language. This finding has implications for computational political science: topic models alone miss the most interesting variation; supervised and lexical methods are needed to capture it.

The Archelec corpus, despite its OCR noise and the formulaic nature of its documents, proves rich enough to support detailed cross-party and temporal analyses. A natural extension would be to bring in transformer-based methods — fine-tuned CamemBERT for classification, or contextual embeddings for studying how the semantics of key political words shift across elections. But the classical methods have carried us further than one might expect, and there is something to be said for that.

References

- Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent Dirichlet Allocation. *Journal of Machine Learning Research*, 3:993–1022.
- Bonnaïfous, S. and Tournier, M. (2003). Analyse du discours, lexicométrie, communication et politique. In Sfez, L., editor, *Dictionnaire critique de la communication*. PUF, Paris.
- Budge, I., Klingemann, H.-D., Volkens, A., Bara, J., and Tanenbaum, E. (2001). *Mapping Policy Preferences: Estimates for Parties, Electors, and Governments 1945–1998*. Oxford University Press.
- Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of ACL*, pages 8440–8451.
- Gaxie, D. (1978). *Le cens caché: inégalités culturelles et ségrégation politique*. Éditions du Seuil, Paris.
- Grimmer, J. and Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297.
- Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.

- Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. In *ECML '98*, pages 137–142. Springer.
- Lee, D. D. and Seung, H. S. (1999). Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791.
- Meguid, B. M. (2005). Competition between unequals: The role of mainstream party strategy in niche party success. *American Political Science Review*, 99(3):347–359.
- Monroe, B. L., Colaresi, M. P., and Quinn, K. M. (2008). Fightin’ words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, 16(4):372–403.
- Quinn, K. M., Monroe, B. L., Colaresi, M., Crespin, M. H., and Radev, D. R. (2010). How to analyze political attention with minimal assumptions and costs. *American Journal of Political Science*, 54(1):209–228.
- Roberts, M. E., Stewart, B. M., Tingley, D., Lucas, C., Leder-Luis, J., Gadarian, S. K., Albertson, B., and Rand, D. G. (2014). Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4):1064–1082.
- Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3):351–357.
- Röder, M., Both, A., and Hinneburg, A. (2015). Exploring the space of topic coherence measures. In *WSDM '15*, pages 399–408. ACM.
- van der Maaten, L. and Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008.

Appendices

A. Profession classification via zero-shot inference

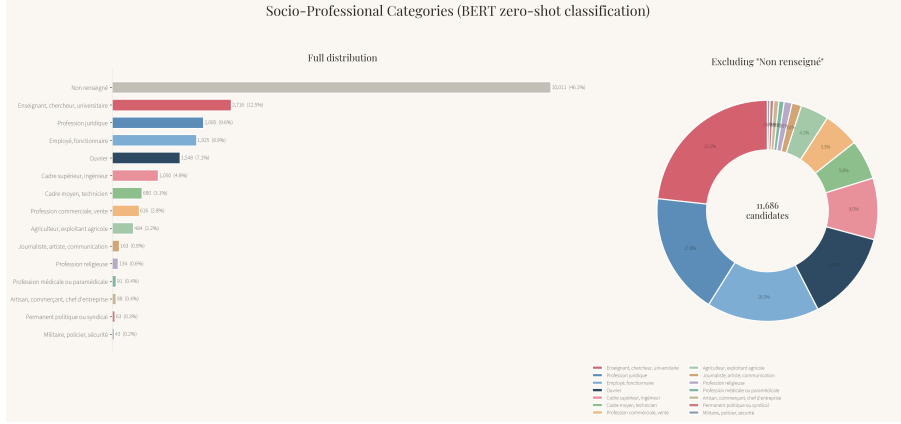


Figure 16: Distribution of BERT-classified PCS categories across the full corpus. Teachers and researchers form the largest group (12.5%), followed by legal professionals (9.6%).

The zero-shot classification pipeline works as follows. For each unique profession string p in the corpus, the XLM-RoBERTa model evaluates the entailment probability of the hypothesis “Cette personne travaille comme c_i ” for each of the 15 candidate labels. The predicted category is $\hat{c} = \arg \max_i P(\text{entailment} | p, c_i)$. This NLI-based approach bypasses the need for training data entirely and handles the high lexical variability of the profession field (abbreviations, regional terms, compound titles) much better than a rule-based approach could.

The resulting distribution (Figure 16) is consistent with the known sociological profile of French political candidates. The strong presence of teachers reflects a long tradition documented in the political science literature; the over-representation of legal professionals echoes the historical dominance of lawyers in parliamentary life.

B. Additional topic model comparisons

We experimented with BERTopic [Grootendorst, 2022] as a third paradigm, using Sentence-Transformers embeddings with UMAP dimensionality reduction and HDBSCAN clustering. While BERTopic produces semantically coherent topic clusters, the resulting topics are less stable across random seeds than LDA, and the framework’s reliance on dense embeddings makes it difficult to perform the same interpretive analyses (coefficient inspection, log-odds computation) that we rely on throughout this paper. We therefore use LDA as our primary model but acknowledge that BERTopic represents a promising direction for future work.

The NMF model, despite achieving higher C_v coherence (0.71 vs 0.49), was rejected because it assigns 84.5% of documents to a single dominant topic. This pathological concentration is a known failure mode of NMF in the absence of the Dirichlet priors that regularise LDA’s topic distributions. NMF excels at isolating hyper-specific micro-clusters (e.g., animal welfare candidates, specific regional issues) but fails to provide a balanced thematic decomposition of the corpus.

C. Topic–metadata cross-analyses

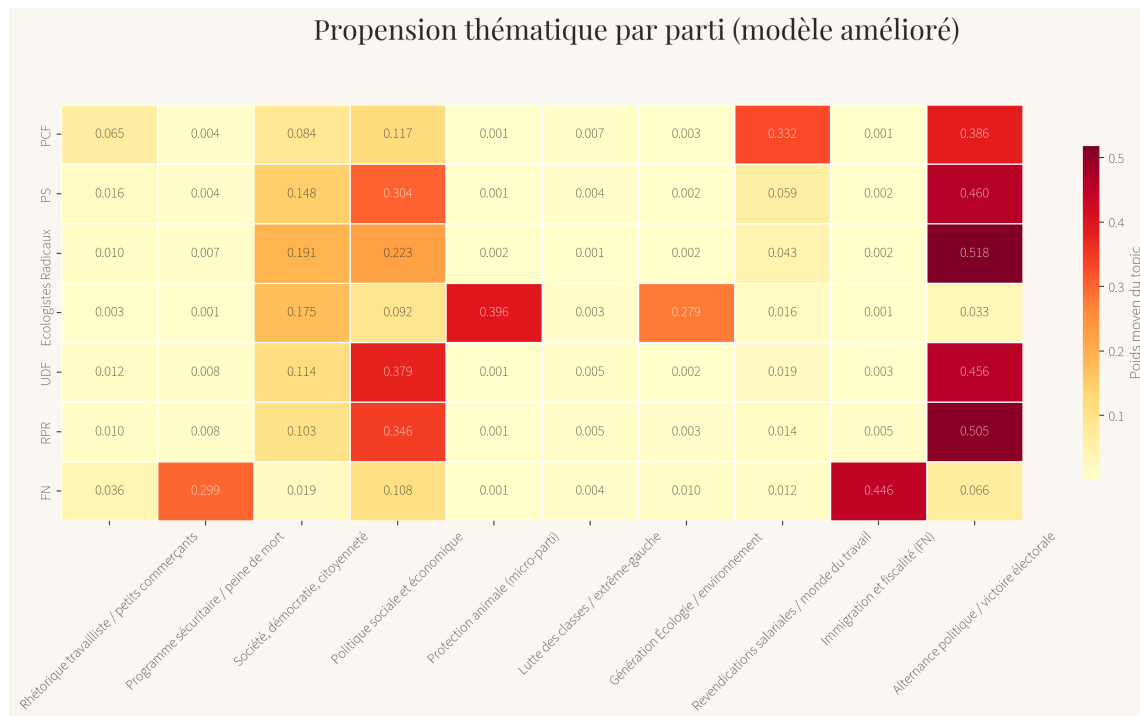


Figure 17: Full heatmap of topic shares by political family. Each column is a topic, each row a family. Darker cells indicate higher share. The diagonal pattern confirms that each family has a distinctive thematic profile, though no topic is exclusive to any one family.

The full heatmap (Figure 17) confirms the specialisation patterns described in Section 5. No topic is exclusive to a single family — the closest to exclusivity is the FN rhetoric topic (Topic 8), which the *Extrême droite* dominates but does not monopolise, since some other families address immigration in passing. The cross-cutting nature of most topics reinforces the central finding that ideological differentiation operates at the vocabulary level, not the topic level.

D. Methodological notes

Reproducibility. All computations are checkpointed. The t-SNE embedding, which is stochastic, is computed once and saved. LDA models are trained with a fixed random seed.

German-language filtering. The Alsace-Moselle departments have a tradition of bilingual electoral materials. Without filtering, these produce a spurious “German language” topic unrelated to French political themes. We identify and remove them using a simple language-detection heuristic applied to the raw text.

OCR quality. The Archelec transcriptions contain recognition errors, particularly in older documents (1973, 1978) where printing quality and typefaces differ from modern standards. Common OCR artefacts include character confusions (*///*, *O/O*), word splits, and header/footer noise. Our preprocessing removes the most systematic artefacts (the “Sciences Po / fonds CEVIPOF” watermark), but residual noise remains. A future improvement would be to apply post-OCR correction using language models.